

Flight Departure Delay Prediction

US domestic flights, 2019 BTS on-time performance data — full results report

Source notebook: `flight_delay_prediction_bts.ipynb` | Train/test: Jan–Oct 2019 / Nov–Dec 2019 (time-based hold-out) | 1,154,661 flights, 5 carriers, 20 airports

This project predicts departure delay for US domestic flights three ways: whether a flight leaves more than 15 minutes late (the D15 metric airlines report on), which of four severity bands it lands in, and the delay in minutes. Almost all of the predictive power sits in two engineered feature groups — **aircraft rotation** (tracking each tail number through its day) and **airport/network congestion**. Dropping either group roughly halves regression R^2 ($0.44 \rightarrow \sim 0.21$); the two groups are complementary, not redundant. Choice of algorithm (XGBoost vs LightGBM) was not found to matter as much as feature engineering.

Results at a glance

Question	Model	Headline result
More than 15 min late?	XGBoost classifier	ROC-AUC 0.843 (0.846 cross-val), top-5% precision 0.977
Which severity band?	LightGBM + ordinal decomposition	macro-F1 0.56, quadratic kappa 0.55–0.57, macro-AUC 0.84
How many minutes?	XGBoost regressor	R^2 0.44, MAE 11.5 min, 83% of errors under 15 min

All figures are on a time-based Nov–Dec hold-out (trained on Jan–Oct), so the same-day congestion state used at test time was never seen during training. The trivial "always on time" baseline scores 0.823 accuracy and 0.226 macro-F1 on this test set — severity numbers should be read against that floor, not against raw accuracy.

Data & target

US BTS *Reporting Carrier On-Time Performance*, 2019, public. Filtered to 5 carriers (WN, DL, AA, OO, UA) and the 20 busiest airports (keeps the network closed, which the rotation features require); cancellations dropped. 1,172,798 rows loaded, 18,137 cancelled/incomplete dropped → **1,154,661 flights**, 3,777 distinct tail numbers, 358 routes.

Delay statistic	Value (min)
Mean	13.1
Median	-1
P75	9
P90	44
P99	217
Max	2058

4-class severity bands

Class	Band	Reason for the cut	Share of flights
0	≤ 15 min	On time, the D15 standard	79.654%
1	15–45 min	Minor, usually absorbed by schedule padding	10.528%
2	45–120 min	Passengers start missing connections	6.779%
3	> 120 min	Crew duty limits and aircraft swaps	3.039%

Binary >15 min rate (whole year): 0.197. Test-period (Nov–Dec) >15 min rate: 0.171.

Feature engineering

Aircraft rotation (11 features)

An aircraft flies several legs a day on the same tail number; if it lands late, the next departure starts late regardless of crew effort. Rebuilt each aircraft's sequence of flights per day via `Tail_Number` + `FlightDate`, using `.shift()` so a flight only sees its own aircraft's earlier legs.

- `prev_arr_delay`, `prev_dep_delay`, `prev2_arr_delay` — delay of prior leg(s)
- `prev_taxi_out`, `sched_turn_min`
- `turn_slack` = scheduled turn – inbound delay (buffer left after a late inbound)
- `leg_num_today`, `legs_remaining`, `tail_legs_today`
- `tail_delay_sofar`, `prev_late15`

Airport / network state (15 features)

Delay propagates through airports independent of aircraft: a hub running late all afternoon tends to stay late. Built from prior flights only, at four levels — origin, destination, network-wide, and carrier×origin.

- `origin_dep_sofar`, `origin_dep_last10`, `origin_frac_late20`
- `dest_dep_sofar`, `dest_frac_late20`
- `net_dep_sofar`, `net_frac_late50`
- `car_org_sofar`, `route_last5`
- `sched_padding` (schedule slack vs historical actual block time)
- `hour_sin/cos`, `doy_sin/cos` cyclical encodings

13 baseline (calendar/route/carrier identity + historical means) + 11 rotation + 15 network features = **81 features** after one-hot encoding origin/destination/carrier.

All history/congestion features are **expanding means over prior flights only**, current row excluded — the same design used consistently for train and test rows, so the model only ever sees information that would genuinely have been available at prediction time.

What the raw pattern looks like

Previous leg's arrival delay	P(this leg > 15 min late)	Mean delay (min)	n
< -15	0.107	5.10	172,169
-15 .. -5	0.115	6.05	161,965
-5 .. 5	0.157	9.99	659,147
5 .. 15	0.227	14.17	54,563
15 .. 30	0.493	23.70	39,382
30 .. 60	0.776	41.71	32,306
60 .. 120	0.863	74.41	21,511
> 120	0.841	150.23	13,618

When the inbound aircraft was on time, ~12% of flights leave late; when it was more than two hours late, over 70% do. `turn_slack` captures the same effect directly: once a late inbound has eaten the scheduled turn, the departure is in trouble regardless of the raw size of the inbound delay.

Train / test split

Train Jan–Oct (967,190 flights), test Nov–Dec (187,471 flights) — a time-based split. A random split would put flights from the same airport on the same afternoon on both sides, letting the model see the congestion state it is meant to predict, which inflates every score and does not match how the model would actually run in production.

Severity classification (4 ordered bands)

Class 0 ("on time") is 82% of the test set, so raw accuracy is nearly meaningless here — always predicting "on time" already scores 0.823 accuracy. Two model families were compared: a direct 4-class LightGBM, and an ordinal decomposition that trains three cumulative binary models ($P(>15)$, $P(>45)$, $P(>120)$), enforces monotonicity, and differences them into class probabilities.

Model	Accuracy	Balanced acc.	Macro F1	Weighted F1	Quad. kappa	Macro AUC
Always "on time" (trivial baseline)	0.8227	0.2500	0.2257	—	0.0000	—
No rotation block (ablation)	0.8286	0.3210	0.3364	0.7645	0.2675	0.7360
LightGBM 4-class (full features)	0.8676	0.4840	0.5571	0.8405	0.5502	0.8430
Ordinal (argmax)	0.8666	0.4761	0.5500	0.8379	0.5382	0.8417
Ordinal (expected band)	0.8509	0.4707	0.5318	0.8353	0.5733	0.8417

Dropping the rotation block costs the most of any single ablation: macro F1 falls from 0.557 to 0.336, kappa from 0.550 to 0.267.

Per-class detail — LightGBM 4-class

Class	Precision	Recall	F1	Support
On time	0.881	0.991	0.933	154,232
Short 15–45	0.713	0.251	0.371	18,611
Medium 45–120	0.682	0.351	0.463	10,581
Long >120	0.703	0.343	0.461	4,047

Per-class detail — ordinal, expected band

Class	Precision	Recall	F1	Support
On time	0.897	0.963	0.929	154,232
Short 15–45	0.423	0.351	0.384	18,611
Medium 45–120	0.659	0.342	0.450	10,581
Long >120	0.922	0.227	0.365	4,047

Precision on the three delay bands runs 0.68–0.71 (direct model) but recall is only 0.25–0.35 — the model stays on "on time" unless confident. Decided by expected band, the ordinal model gives the best kappa (0.573) and a sharper high-severity operating point: 0.922 precision on the "long" band at 0.227 recall, against 0.703 precision at 0.343 recall for the direct model. Which model is "better" depends on whether the use case wants broad recall or high-confidence flags.

Binary triage (> 15 min late)

Model	ROC-AUC	PR-AUC	Precision @5%	Precision @15%	Recall @15%
No rotation block (ablation)	0.7406	0.4561	0.691	0.452	0.397
No network block (ablation)	0.7847	0.5984	0.970	0.547	0.480
Full feature set	0.8432	0.6840	0.977	0.641	0.562

Dropping rotation costs 0.103 ROC-AUC; dropping network costs 0.058 ROC-AUC — rotation carries more standalone signal for this task, but both blocks contribute independently. Base rate (test set): 0.171.

Operating points (rank-based flagging)

Flag top	Flights flagged	Precision	Recall	Lift vs base rate
2%	3,749	1.000	0.117	5.85×
5%	9,373	0.977	0.286	5.72×
10%	18,747	0.797	0.466	4.66×
15%	28,120	0.641	0.562	3.75×
20%	37,494	0.539	0.631	3.15×
30%	56,241	0.418	0.734	2.45×

Top-2% precision of 100% is partly mechanical: a plane already hours late will very likely leave late again. The more informative operating range is the middle of the risk distribution, where precision runs 0.54–0.64.

Regression (delay in minutes)

Target clipped to $[-30, 180]$ so a small number of multi-hour outliers don't dominate squared error. Four runs: leave-one-out ablations (no rotation, no network), the full feature set, and a two-stage hurdle model ($P(\text{late}) \times E[\text{min} \mid \text{late}] + P(\text{on time}) \times E[\text{min} \mid \text{on time}]$).

Model	R^2	RMSE	MAE	Within 15 min	Within 30 min
No rotation block	0.2098	26.84	15.15	75.0%	90.2%
No network block	0.2069	26.89	17.03	62.6%	85.7%
Full feature set	0.4441	22.51	11.50	83.1%	93.1%
Two-stage hurdle	0.4456	22.48	11.41	83.4%	93.2%

Dropping either feature block costs about the same (rotation: $-0.234 R^2$, network: $-0.237 R^2$), and each alone recovers less than half of what the two give together — they are complementary, not redundant. The two-stage hurdle model only adds $+0.0015 R^2$ over the plain regressor, suggesting the limiting factor at this point is the data (no weather, no crew/ATC information), not the model.

Feature importance

Rotation features contribute **41.2%** of total gain from just 11 of 81 features; network features contribute **28.2%** from 15 of 81. Together, 26 of 81 features (32%) account for ~69% of total model gain.

Feature	Block	Gain share
turn_slack	Rotation	0.1793
prev_arr_delay	Rotation	0.1168
origin_frac_late20	Network	0.0924
dest_frac_late20	Network	0.0822
sched_turn_min	Rotation	0.0403
prev_dep_delay	Rotation	0.0257
car_org_sofar	Network	0.0232
leg_num_today	Rotation	0.0217
Reporting_Airline_00	Baseline	0.0170
route_last5	Network	0.0161

SHAP values (20,000-row sample of the test set) confirm the same ranking and direction of effect as the gain-based importances above.

Validation

1. Rolling-origin cross-validation (3 expanding folds inside the training window)

Fold ROC-AUCs: 0.848, 0.844, 0.846 — mean **0.8460 ± 0.0016**. Consistent with the single Nov–Dec hold-out score of 0.843 — that result is not a lucky window.

2. Calibration

Brier score: **0.0883**. Predicted → observed frequency across 8 quantile bins: 0.03→0.02, 0.04→0.04, 0.06→0.06, 0.08→0.07, 0.11→0.10, 0.15→0.14, 0.24→0.23, 0.70→0.71 — predicted probabilities track observed frequencies closely across the whole range.

3. Permutation test

Labels shuffled and the model refit 5 times: shuffled-label ROC-AUC **0.488 ± 0.040** (i.e. chance) versus the real model's **0.843** — **9 standard deviations above chance**.

4. Learning curve

Training rows	Train AUC	Test AUC	Train–test gap
96,719	0.898	0.832	0.066
241,797	0.876	0.838	0.038
483,595	0.867	0.840	0.027
967,190	0.860	0.842	0.018

The train/test gap shrinks steadily as training data grows (0.066 → 0.018), indicating the model is not badly overfit and is not obviously data-starved at the current sample size.

From score to decision: cost-based thresholds

A risk score alone isn't a decision — that requires the cost ratio between a false alarm (watching a flight that turns out fine) and a miss (a late flight with no warning). For a miss costing C times a false alarm, the optimal threshold minimises $FP + C \times FN$.

Cost ratio (miss : false alarm)	Threshold	Flagged	Precision	Recall	F1
2:1	0.35	11.7%	0.735	0.504	0.598
4:1	0.20	22.0%	0.509	0.655	0.573
8:1	0.12	39.0%	0.349	0.797	0.486

The low recall seen in the severity/binary tables above is a business choice about acceptable false-alarm rate, not an inherent property of the model — it moves along this table as the cost ratio changes.

Limitations

- **No weather.** BTS weather fields are assigned after the fact and would leak; a real forecast feed is the obvious next addition.
- **Prediction horizon is 1–3 hours** — `prev_arr_delay` needs the inbound leg to have already landed. Day-ahead prediction would need to chain leg by leg and would score lower.
- **Top-2% precision (100%) is partly mechanical:** a plane already hours late will leave late again. The useful range for a real triage system is the middle of the risk distribution, where precision runs 0.54–0.64.
- **Scope:** 2019 only, 20 airports, cancellations excluded — results should not be assumed to transfer to other years, regional airports, or the post-2020 network.

Summary

Task	Metric	Result	Reference
Severity (4-class)	Balanced accuracy	0.48	0.25 random
	Macro F1	0.56	0.23 trivial
	Quadratic kappa	0.55 (0.57 ordinal)	0.00 trivial
	Macro OvR AUC	0.84	–
Binary (>15 min)	ROC-AUC	0.843	–
	ROC-AUC, rolling CV	0.846 ± 0.002	–
	Precision @ top 5%	0.977	0.171 base rate (5.7×)
	Brier score	0.088	–
Regression (minutes)	R ²	0.444	–
	MAE	11.5 min	–
	Within 15 min	83%	–

Feature block removed	Regression R ²	Binary AUC	Severity macro F1
Rotation	0.210	0.741	0.336
Network	0.207	0.785	–
Nothing (full model)	0.444	0.843	0.557

Generated from `flight_delay_prediction_bts.ipynb` — all figures above come from a single top-to-bottom run of that notebook against the 2019 BTS on-time performance extract (5 carriers, 20 airports, 1,154,661 flights). See the repository README and notebook for full code, plots, and methodology detail.